ABHAY DESHMUKH

Software / ML Developer & ECE Engineer
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PROFESSIONAL SUMMARY

Final-year Electronics & Communication Engineering student with an 8.7 CGPA and hands-on experience building Python/ML applications, Streamlit interfaces and embedded/IoT systems. Experienced in developing end-to-end projects involving machine-learning inference, sensor-data pipelines, AWS IoT Core, MQTT, CNNs and data preprocessing. Strong foundation in programming, problem solving and software development, with additional industrial automation experience.

TECHNICAL SKILLS

Languages: Python, C, Embedded C
Software & Web: HTML5, CSS3, Streamlit, GitHub
Machine Learning: ANN, CNN, Classification, Model Inference, Data Preprocessing, NumPy, Pandas
Cloud / IoT: AWS IoT Core, MQTT, ESP32, Sensor Integration
Automation: PLC Programming, Ladder Logic, SCADA/HMI, Modbus, OPC UA, MQTT

PROJECTS

IoT-Enabled Predictive Maintenance System github.com/abhay178125

- ESP32-based condition-monitoring system using temperature, vibration (ADXL345), pressure (MPX5700AP), humidity and noise sensors.
- Streamed sensor data to AWS IoT Core over MQTT and evaluated ANN and ensemble ML models for machine-fault classification; ANN achieved the best classification accuracy.
- Designed the system as an end-to-end monitoring pipeline covering sensing, cloud ingestion and predictive fault detection.

Tech: Python, Machine Learning, ANN, ESP32, AWS IoT Core, MQTT, Embedded C

Automatic Modulation Classification using 1-D CNN github.com/abhay178125/snr-2

- Built a Streamlit application that generates noisy I/Q signals and classifies BPSK, QPSK, 16-QAM, 64-QAM, CPFSK and WBFM.
- Implemented CNN inference with a TensorFlow/Keras path and a pure-NumPy fallback, with model caching for deployment.
- Added interactive time-domain, constellation and probability visualizations using Plotly.

Tech: Python, CNN, NumPy, TensorFlow/Keras, Streamlit, Plotly

Supply Chain Risk Classification App github.com/abhay178125/supply-chain

- Developed a Streamlit prediction application using a trained classification model and feature scaler to classify supply-chain risk.
- Built an interactive 23-feature input workflow covering logistics, inventory, transportation, IoT, supplier and disruption indicators.
- Persisted the trained model and scaler with Joblib and integrated preprocessing directly into the prediction flow.

Tech: Python, Pandas, Scikit-learn/ML, Streamlit, Joblib

CERTIFICATIONS

- AWS Certification
- Python Certification
- IoT Certification
- MATLAB Certification

PARTICIPATION

- Participated in Engineer's Day Celebration 2026.
- Participated in EngineerX – Mega Project Expo.
- Participated in TechCanvas – Technical Poster Competition.
- Participated in Industry Arena – Industry Simulation & Optimization Competition.

EDUCATION

B.Tech, Electronics & Communication Engineering — Tulsiramji Gaikwad-Patil College of Engineering and Technology (TGPCET), Nagpur | April 2027
CGPA: 8.7 (Final Year)
HSC (12th), Jawahar Navodaya Vidyalaya, Yavatmal: 83.60% | **SSC (10th):** 89.60%